INT374 : DATA ANALYTICS WITH POWER BI

PROJECT REPORT

(Project Semester August-January 2025)

PMGSY – PHYSICAL AND FINANCIAL PERFORMANCE ANALYSIS

Discipline of CSE
Lovely School of Computer Science & Engineering
Lovely Professional University, Phagwara

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CERTIFICATE

This is to certify that Hasith Mutha bearing Registration No. 12301431 has successfully completed the course project titled **"PMGSY – Physical and Financial Performance Analysis Using Power BI"** under my guidance and supervision for the course INT234. The work presented is original and meets the academic requirements.

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DECLARATION

I, Hasith Mutha, student of B.Tech Computer Science & Engineering under CSE Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine. This project work is original and has not been submitted elsewhere for any academic award.

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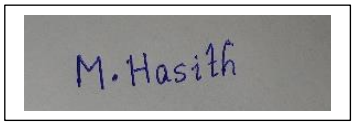

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1. INTRODUCTION

Rural road infrastructure is one of the most important foundations for the overall development of any country. Well-connected rural roads play a crucial role in improving the quality of life of rural communities by providing easier access to essential services such as education, healthcare, markets, and employment opportunities. Good connectivity also reduces travel time and transportation costs, enabling people to move goods and services more efficiently. In the agricultural sector, improved road networks allow farmers to transport crops to markets quickly, reduce post-harvest losses, and increase their income. In a developing country like India, strengthening rural road infrastructure is essential for reducing regional inequality and promoting inclusive and balanced economic growth.

The Pradhan Mantri Gram Sadak Yojana (PMGSY) is a flagship rural development scheme launched by the Government of India with the objective of providing all-weather road connectivity to unconnected rural habitations. Since its introduction, PMGSY has made a significant contribution to improving rural infrastructure across different states and regions of the country. The scheme has helped enhance mobility, improved access to public services, and supported socio-economic development in rural areas. Over the years, PMGSY has evolved through multiple phases such as PMGSY-I, PMGSY-II, and PMGSY-III. In addition, special schemes like RCPLWEA and PM-JANMAN were introduced to address connectivity challenges in remote, tribal, and difficult geographical regions.

Due to the large scale of PMGSY implementation, which involves thousands of road construction projects, bridge works, and substantial public investment, effective monitoring and evaluation are extremely important. Monitoring such large infrastructure programs using traditional reporting methods, such as static tables and manual reports, often makes it difficult to understand overall performance. These methods fail to provide an integrated view of physical progress and financial utilization, which is necessary for timely decision-making and effective governance.

To address these challenges, there is a growing need for data-driven and visual analytical tools that can present complex information in a simple, clear, and interactive manner. In this project, Power BI is used as a business intelligence and data visualization tool to analyze PMGSY data obtained from official government sources. The project focuses on transforming raw and complex datasets

into interactive dashboards that clearly display key indicators such as road length completion, expenditure utilization, scheme-wise performance, and state-wise progress.

The dashboards developed in this project enable stakeholders, planners, and policymakers to quickly identify trends, compare performance across regions, and detect areas that require improvement. By providing accurate, timely, and visually meaningful insights, this project demonstrates how data analytics and visualization can support better planning, monitoring, and decision-making in large-scale government infrastructure programs like PMGSY.

2. BACKGROUND OF PMGSY

PMGSY was launched by the Government of India in December 2000 with the primary objective of improving rural road connectivity across the country. The scheme was designed to provide all-weather road access to unconnected rural habitations, thereby supporting social and economic development in rural areas. Since its launch, PMGSY has played a significant role in strengthening rural infrastructure and improving accessibility.

Over time, the program was expanded through multiple phases such as PMGSY-I, PMGSY-II, PMGSY-III, RCPLWEA, and PM-JANMAN to address different regional and developmental challenges. Each of these schemes differs in terms of scope, funding pattern, target regions, and implementation strategy. While some schemes focus on providing basic connectivity, others aim at upgrading existing road networks or improving connectivity in remote and difficult regions.

Understanding the performance of these schemes requires an integrated analysis of both physical and financial indicators. Key factors such as sanctioned road length, completion status, expenditure utilization, and bridge construction must be evaluated together to assess overall effectiveness. Such an integrated approach helps in identifying strengths, gaps, and areas that require improved planning and monitoring.

3. OBJECTIVES OF THE PROJECT

The key objectives of this project are:

- To monitor physical progress of PMGSY road projects across India
- To analyze state-wise and scheme-wise financial performance
- To evaluate road length completion and expenditure utilization
- To identify high-performing and low-performing states
- To provide an interactive decision-support dashboard for planners and policymakers

4. SOURCE OF DATASET

The dataset used in this project was collected from **data.gov.in**, the official open data portal of the Government of India. This portal provides authentic, verified, and regularly updated datasets published by various government ministries and departments. Using data from an official government source ensures the accuracy, reliability, and credibility of the analysis carried out in this project.

The dataset specifically relates to the **Pradhan Mantri Gram Sadak Yojana (PMGSY)** and contains detailed information about rural road infrastructure projects implemented across different states and districts in India. It covers multiple PMGSY schemes and provides both physical and financial data required to evaluate project performance.

Link: https://www.data.gov.in/catalog/pradhan-mantri-gram-sadak-yojna-pmgsy

The key attributes included in the dataset are as follows:

- State and district names, which enable geographical and regional analysis
- PMGSY scheme categories such as PMGSY-I, PMGSY-II, PMGSY-III, RCPLWEA, and PM-JANMAN
- Sanctioned road length and completed road length, used to measure physical progress
- Expenditure incurred, recorded in lakhs, to assess financial utilization
- Cost per kilometer, which helps in evaluating cost efficiency across states
- Number of bridges sanctioned and completed, reflecting infrastructure development in difficult terrains

The dataset contains more than 2,000 records, which provides sufficient volume and variety for meaningful analysis. This large data size allows reliable comparison across states, districts, and schemes, and supports the identification of trends, patterns, and performance gaps.

5. TOOLS AND TECHNOLOGIES USED

- **Power BI Desktop:** Power BI Desktop was used as the primary tool for this project to create interactive dashboards and visual reports. It helped in presenting complex PMGSY data in a clear and visually meaningful manner using charts, maps, tables, and KPI cards. Power BI also enabled easy comparison of physical and financial performance across states and schemes.
- **Power Query:** Power Query was used for data cleaning and transformation before analysis. It helped in handling missing values, removing duplicate records, standardizing

column names, and converting data into the required format. This step ensured that the dataset was accurate, consistent, and ready for analysis.

- **DAX (Data Analysis Expressions):** DAX was used to create calculated measures and KPIs required for the dashboard. It helped in computing metrics such as completion percentage, expenditure utilization, cost per kilometer, and aggregated totals. DAX made the dashboard dynamic and responsive to filters and slicers.

6. EXPLORATORY DATA ANALYSIS (EDA) PROCESS

Exploratory Data Analysis (EDA) is a critical phase in any data analytics project, as it helps in understanding the dataset before applying advanced analysis or visualization techniques. The primary purpose of EDA is to explore the data structure, identify patterns, detect inconsistencies, and ensure data quality. In the context of government infrastructure projects such as PMGSY, accurate data analysis is especially important because decisions based on incorrect or incomplete data can lead to ineffective planning and resource allocation.

In this project, EDA was carried out on PMGSY data collected from official government sources. The dataset contained information related to road construction, bridge development, sanctioned costs, expenditure incurred, and scheme-wise implementation across multiple states and districts. Due to the large size and complexity of the dataset, a systematic EDA process was essential to prepare the data for meaningful analysis in Power BI.

6.1 Understanding the Dataset Structure

The initial step in the EDA process involved understanding the overall structure of the dataset. This included reviewing the number of records, number of columns, data types, and the nature of each variable. Columns were classified into categorical variables (such as state name, district name, and scheme type) and numerical variables (such as road length, expenditure, and cost per kilometer).

This step helped in gaining a clear understanding of how the data was organized and which fields were relevant for physical and financial performance analysis. It also helped in identifying unnecessary or redundant columns that could be removed to simplify the data model.

6.2 Handling Missing and Inconsistent Values

One of the major challenges observed in the dataset was the presence of missing and inconsistent values. Some records had missing entries for completed road length, expenditure incurred, or bridge counts. In other cases, inconsistent values were found due to data entry variations or incomplete reporting.

To address this issue, missing values were carefully examined to understand their impact on analysis. Records with critical missing information that could significantly distort results were either corrected (where possible) or excluded from the analysis. For non-critical missing values, appropriate handling techniques were applied to maintain data integrity. This step ensured that calculations such as completion percentage and expenditure utilization were accurate and reliable.

6.3 Standardization of Categorical Fields

Categorical fields such as state names, district names, and scheme categories showed variations in spelling, capitalization, and naming conventions. For example, the same state might appear in different formats across records. Such inconsistencies can lead to incorrect grouping and aggregation in Power BI.

To resolve this, categorical fields were standardized by correcting spelling errors, unifying naming formats, and applying consistent capitalization. Scheme names such as PMGSY-I, PMGSY-II, PMGSY-III, RCPLWEA, and PM-JANMAN were also standardized to ensure accurate scheme-wise comparison. This step was crucial for enabling reliable filtering, grouping, and state-wise analysis in the dashboard.

6.4 Removal of Duplicate Records

The dataset was examined for duplicate records that may have occurred due to repeated data entries or data collection issues. Duplicate records can significantly affect analysis by inflating values such as total road length, expenditure, or number of sanctioned works.

Using Power Query, duplicate rows were identified based on key fields such as state, district, scheme, and project identifiers. These duplicate entries were removed carefully to ensure that each project record was counted only once. This helped in maintaining the accuracy of aggregated metrics and KPIs displayed in the dashboard.

6.5 Data Type Validation and Formatting

Another important step in the EDA process was validating and correcting data types. Some numerical columns were initially stored as text, which could cause errors during calculations. Columns such as sanctioned cost, expenditure incurred, road length, and cost per kilometer were converted to appropriate numerical data types.

Similarly, units of measurement were verified to ensure consistency across records. Financial values were standardized to lakhs, and road length values were checked to ensure uniform units. This step was essential for accurate calculation of totals, averages, and percentages in Power BI.

6.6 Creation of Derived Metrics

After cleaning and validating the dataset, derived metrics were created to support meaningful analysis. These metrics were not directly available in the raw data but were necessary for evaluating performance.

Important derived fields included:

- **Road Length Completion Percentage**, calculated using sanctioned and completed road length
- **Expenditure Utilization Percentage**, calculated using sanctioned cost and expenditure incurred
- **Average Cost per Kilometer**, used to evaluate cost efficiency across states and schemes

These metrics were created using calculated columns and DAX measures. They allowed direct comparison of physical and financial performance across different regions and schemes.

6.7 Outlier Detection and Validation

The dataset was further analyzed to identify unusually high or low values in numerical fields such as cost per kilometer, expenditure, and road length. Such outliers could indicate data entry errors or exceptional real-world conditions.

Outliers were carefully reviewed rather than automatically removed. In many cases, high cost per kilometer values were found in hilly or remote regions, which are expected due to difficult terrain. This step helped distinguish between genuine regional variations and potential data errors.

6.8 Final Data Validation for Dashboard Readiness

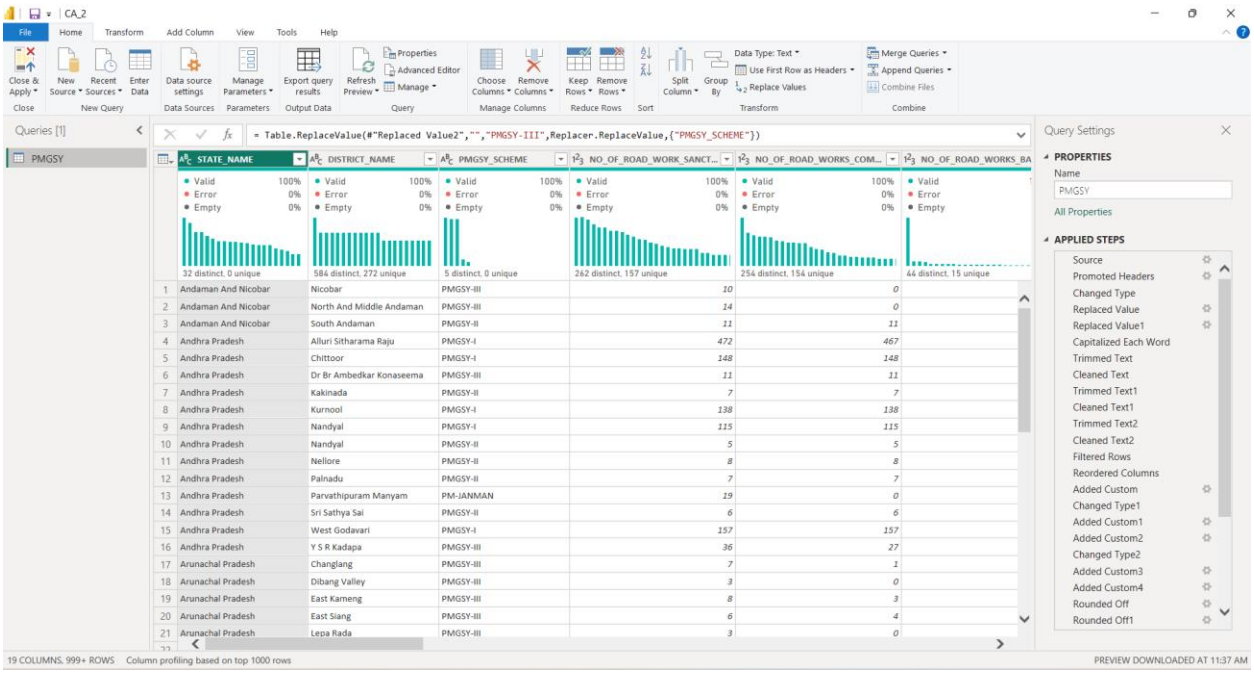
Before loading the data into Power BI visuals, a final validation was performed to ensure consistency across all tables and relationships. Aggregated values were cross-checked, and sample records were manually verified to confirm correctness.

This final review ensured that the dataset was clean, consistent, and analysis-ready, making it suitable for building interactive dashboards and deriving reliable insights.

6.9 Outcome of the EDA Process

The Exploratory Data Analysis process played a crucial role in the success of this project. By systematically cleaning, transforming, and validating the PMGSY dataset, EDA ensured that the insights generated through the Power BI dashboard were accurate and meaningful.

Overall, EDA laid a strong foundation for analyzing the physical and financial performance of PMGSY projects. It enabled effective visualization, supported accurate KPI calculation, and ensured that the conclusions drawn from the dashboard were trustworthy and useful for decision-making.



7. ANALYSIS ON DATASET

7.1 Physical Progress Analysis

Introduction:

Physical progress analysis focuses on evaluating the actual execution of PMGSY road projects across different states in India. While financial allocation is important, the real success of any infrastructure program depends on how effectively the sanctioned works are completed on the ground. This analysis helps in understanding whether approved projects are translating into completed roads and bridges within a reasonable time frame.

In the context of PMGSY, physical progress is a key indicator of implementation efficiency. By examining physical parameters such as road length completion and bridge construction, it becomes possible to compare performance across states and identify regions that are progressing well as well as those facing challenges. This systematic evaluation supports better monitoring and helps authorities focus on improving execution in lagging areas.

Metrics Used

To assess the physical progress of PMGSY projects, the following key metrics were used in the analysis:

- **Total Sanctioned Works**:
  This metric represents the total number of road projects approved under PMGSY for each state. It reflects the scale of infrastructure development planned in a particular region.

- **Completed Road Length**:
  This indicates the actual length of roads that have been constructed and completed. It provides a direct measure of on-ground execution.
- **Length Completion Percentage**:
  This metric measures the proportion of completed road length in comparison to the sanctioned road length. It is a critical indicator of how much of the planned work has been successfully executed.
- **Number of Bridges Completed**:
  Bridge construction is especially important in regions with rivers, uneven terrain, or seasonal accessibility issues. This metric highlights infrastructure development in geographically challenging areas.

Together, these metrics provide a comprehensive view of physical execution efficiency at the state level and allow meaningful comparison across regions.

Formula Used

The **Length Completion Percentage** was calculated using the following formula:

$$\text{Length Completion \%} = \left(\frac{\text{Completed Road Length}}{\text{Sanctioned Road Length}}\right) \times 100$$

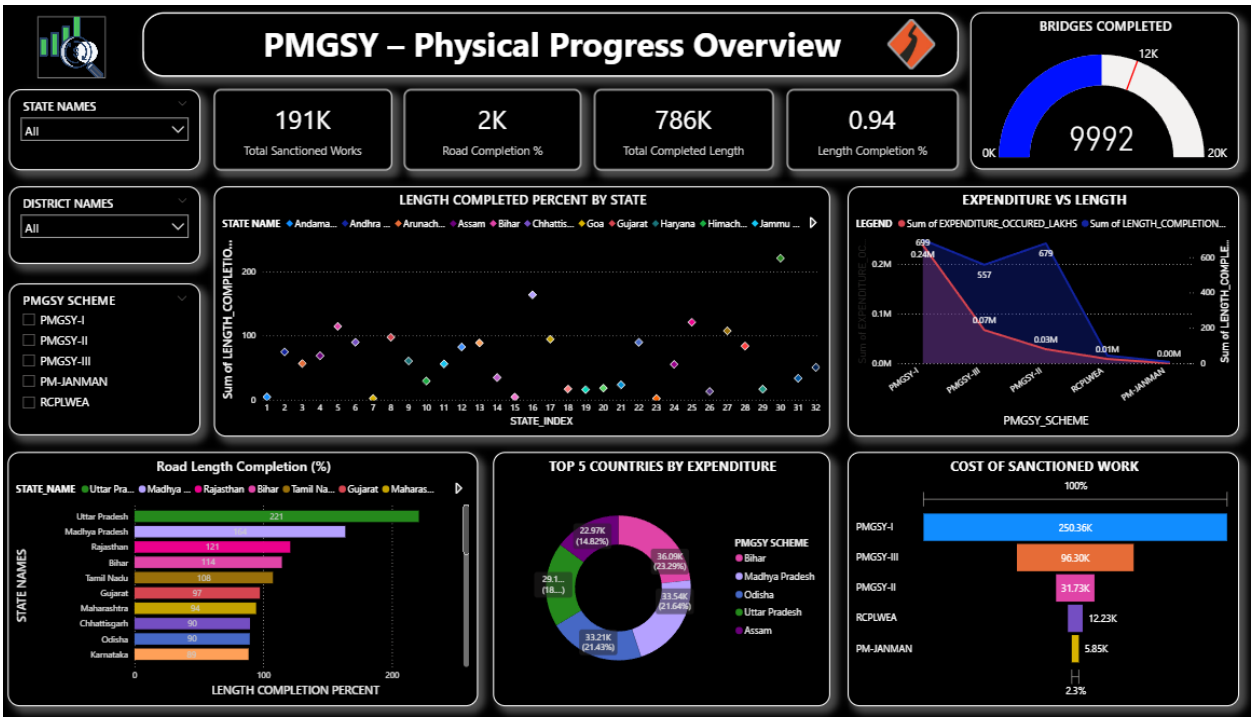
This formula was implemented using **DAX (Data Analysis Expressions)** in Power BI. The use of DAX allows the completion percentage to be calculated dynamically, meaning that the values automatically update based on user-selected filters such as state, scheme, or year.

Analysis Results

The physical progress analysis revealed significant variation in performance across states. Some states demonstrated high road length completion percentages, indicating efficient project execution and timely completion of sanctioned works. These states reflect better coordination, planning, and on-ground implementation.

On the other hand, a few states showed lower completion percentages despite having a large number of sanctioned projects. This suggests the presence of challenges such as funding delays, difficult geographical conditions, administrative bottlenecks, or implementation inefficiencies. In such cases, high project approval does not necessarily translate into completed infrastructure.

Overall, this analysis helps in identifying high-performing states as well as regions that require focused attention and improved monitoring. By highlighting gaps between sanctioned works and completed outputs, physical progress analysis plays a crucial role in supporting better planning, supervision, and execution of PMGSY projects.



7.2 Financial Performance Analysis

Introduction:

Financial performance analysis focuses on evaluating how effectively the allocated funds under PMGSY are utilized across different schemes and states. Since PMGSY involves large public investment, efficient financial management is essential to ensure that infrastructure projects are completed on time and within budget. Proper utilization of funds directly impacts the pace and quality of road construction.

In large-scale government programs like PMGSY, merely sanctioning a high budget does not guarantee successful project implementation. Therefore, it is important to analyze expenditure patterns, utilization levels, and cost efficiency to understand the overall financial health of the program. This section examines key financial indicators to assess whether funds are being used optimally across states and schemes.

Metrics Used

To evaluate financial efficiency under PMGSY, the following key metrics were used in the analysis:

- **Sanctioned Cost**:
  This represents the total budget approved for PMGSY projects. It indicates the level of financial commitment made by the government for rural road development across different schemes and regions.

- **Expenditure Occurred**:
  This metric shows the actual amount of money spent on project execution. Comparing expenditure with sanctioned cost helps in understanding how much of the allocated budget has been utilized.
- **Cost per Kilometer**:
  This measures the average cost required to construct one kilometer of road. It is an important indicator of cost efficiency and helps in comparing construction costs across different states and geographical regions.

Together, these metrics provide a comprehensive view of financial performance and help identify differences in spending efficiency across schemes and states.

Formula Used

The **Expenditure Utilization Percentage** was calculated using the following formula:

$$\text{Expenditure Utilization \%} = \left(\frac{\text{Expenditure Occurred}}{\text{Sanctioned Cost}}\right) \times 100$$

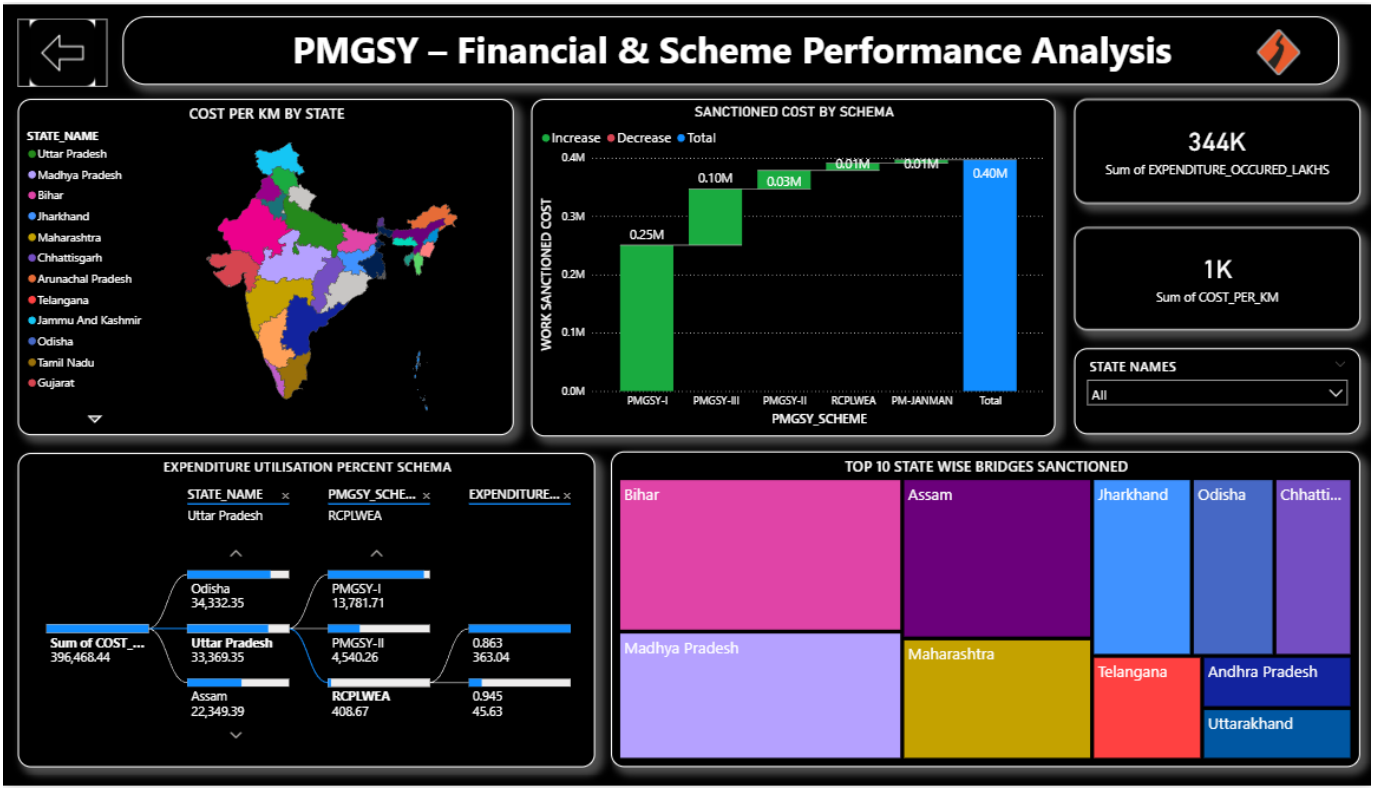
This calculation was implemented using **DAX measures** in Power BI. The use of DAX allows expenditure utilization to be calculated dynamically, enabling users to analyze financial performance based on selected states, schemes, and time periods through slicers.

Analysis Results

The financial performance analysis reveals notable differences in expenditure utilization across schemes and states. In several cases, schemes and states with high sanctioned budgets show relatively low expenditure utilization percentages. This indicates potential implementation challenges such as project delays, administrative bottlenecks, contractor-related issues, or inefficient planning.

On the other hand, some states and regions demonstrate efficient fund utilization, where expenditure closely matches the sanctioned cost. Such regions reflect better financial management, effective monitoring, and stronger execution capacity. The analysis of cost per kilometer further highlights regional variations, with higher costs observed in hilly and remote areas due to terrain and accessibility challenges.

Overall, financial performance analysis helps in identifying schemes and states where financial planning and monitoring need improvement. By highlighting gaps between sanctioned funds and actual expenditure, this analysis supports better allocation of public funds, improved budget control, and more effective implementation of PMGSY projects.



7.3 Scheme-wise Analysis

Introduction

Scheme-wise analysis is carried out to understand the contribution, focus, and performance of different PMGSY schemes. Since PMGSY has been implemented in multiple phases with different objectives, analyzing each scheme separately helps in identifying how government priorities have evolved over time and how resources are distributed across schemes.

Each PMGSY scheme differs in terms of scope, target regions, funding pattern, and implementation strategy. Therefore, scheme-wise analysis provides clarity on which schemes contribute most to rural road development and how efficiently funds are utilized under each scheme.

Key Aspects Considered

The following aspects were considered while performing scheme-wise analysis:

- Total sanctioned cost under each scheme
- Expenditure occurred by scheme
- Road length sanctioned and completed

- Relative contribution of each scheme to overall PMGSY implementation

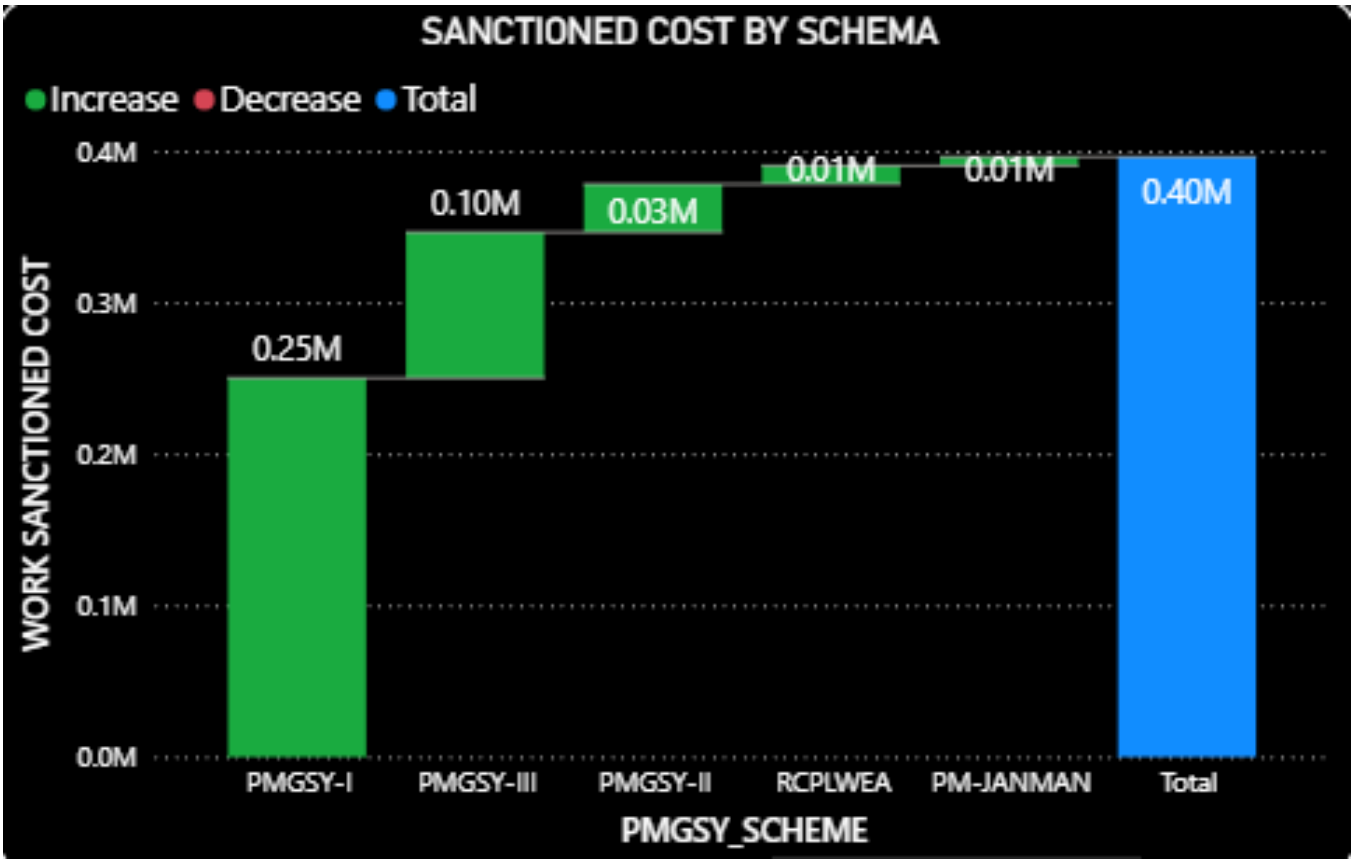
These aspects help in comparing schemes not only in terms of investment but also in execution performance.

Analysis Results

The analysis clearly shows that PMGSY-I and PMGSY-III account for the largest share of infrastructure investment. PMGSY-I primarily focused on providing basic rural connectivity, while PMGSY-III emphasizes upgradation and consolidation of existing rural road networks. Due to their wider scope and coverage, these schemes dominate both sanctioned cost and road length.

In contrast, newer and region-specific schemes such as RCPLWEA and PM-JANMAN contribute a smaller share of total investment. However, their role is critical in improving connectivity in remote, tribal, and geographically challenging regions. Although financial allocation is lower, the impact of these schemes is significant for inclusive development.

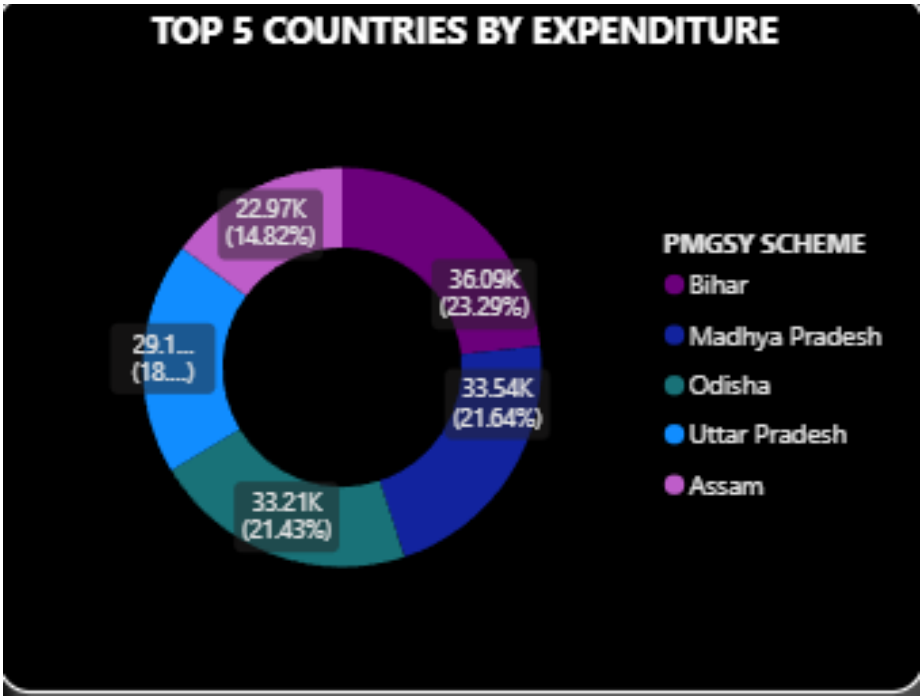
The scheme-wise expenditure pattern also highlights differences in utilization efficiency. Some schemes show higher utilization rates, while others indicate slower progress, suggesting the need for improved monitoring and execution strategies.



Interpretation and Importance

Scheme-wise analysis helps policymakers understand whether investments are aligned with developmental goals. High allocation to PMGSY-I and PMGSY-III reflects a strong focus on expanding and strengthening rural road infrastructure. Meanwhile, targeted schemes demonstrate the government's intention to address regional inequalities.

This analysis also helps in identifying schemes that may require additional support, policy revision, or improved execution mechanisms. By comparing performance across schemes, authorities can ensure balanced development and effective utilization of public funds.



7.4 State-wise Comparative Analysis

Introduction

State-wise comparative analysis is essential for understanding regional differences in PMGSY implementation. Since states vary in population size, geographical conditions, and development needs, comparing states helps in identifying disparities in infrastructure development and execution efficiency.

This analysis evaluates how different states perform in terms of sanctioned works, expenditure, road length completion, and cost efficiency.

Key Metrics Considered

The following parameters were used for state-wise comparison:

- Total sanctioned cost by state

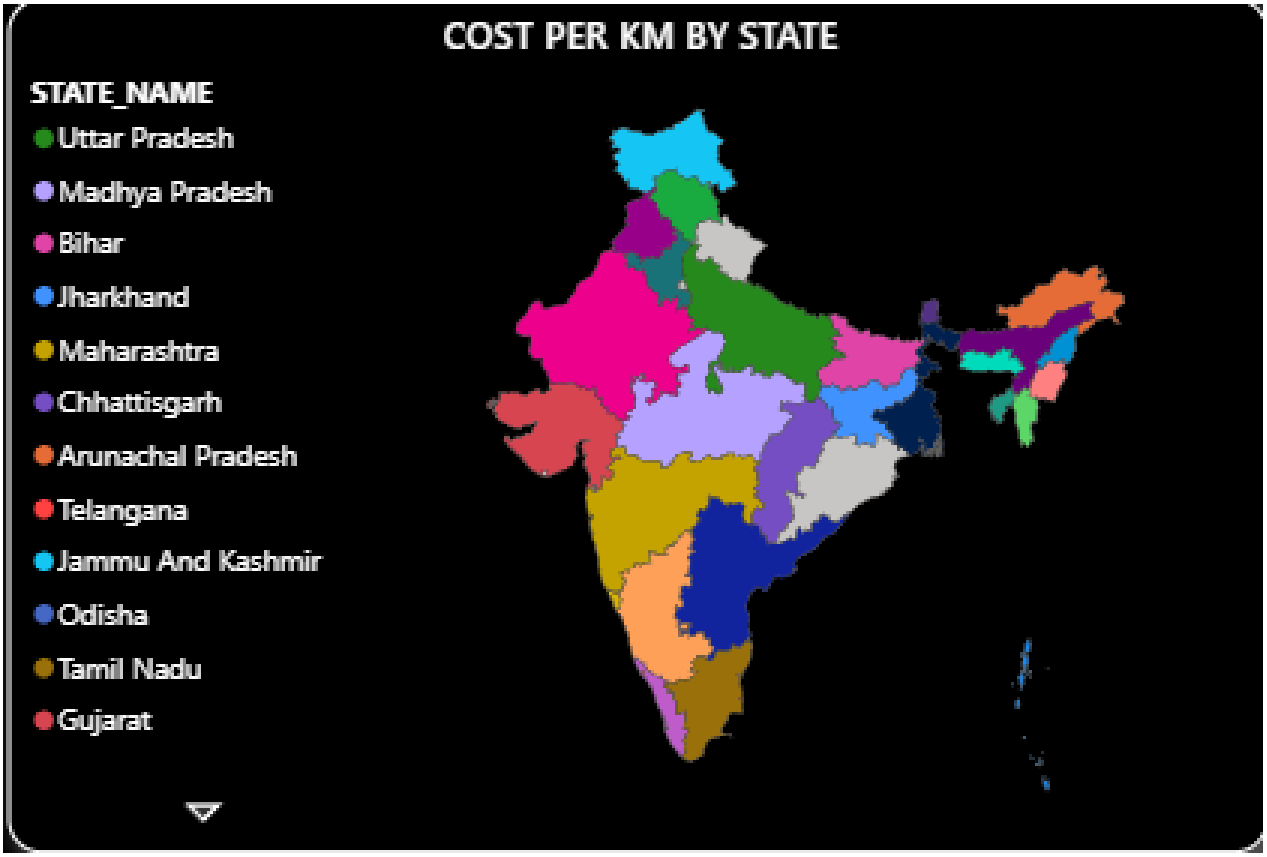
- Expenditure occurred by state
- Road length completed
- Cost per kilometer
- Number of bridges constructed

These metrics together provide a comprehensive view of state-level performance.

Analysis Results

The analysis indicates that states such as Uttar Pradesh, Madhya Pradesh, and Bihar dominate in total expenditure and road length. These states have large rural populations and extensive road networks, which results in higher allocation of funds and a larger number of sanctioned projects.

On the other hand, northeastern and hilly states show a higher cost per kilometer. This is mainly due to difficult terrain, forest cover, and limited accessibility. Although these states receive lower total funds, the higher cost per kilometer reflects the complexity of construction in such regions.



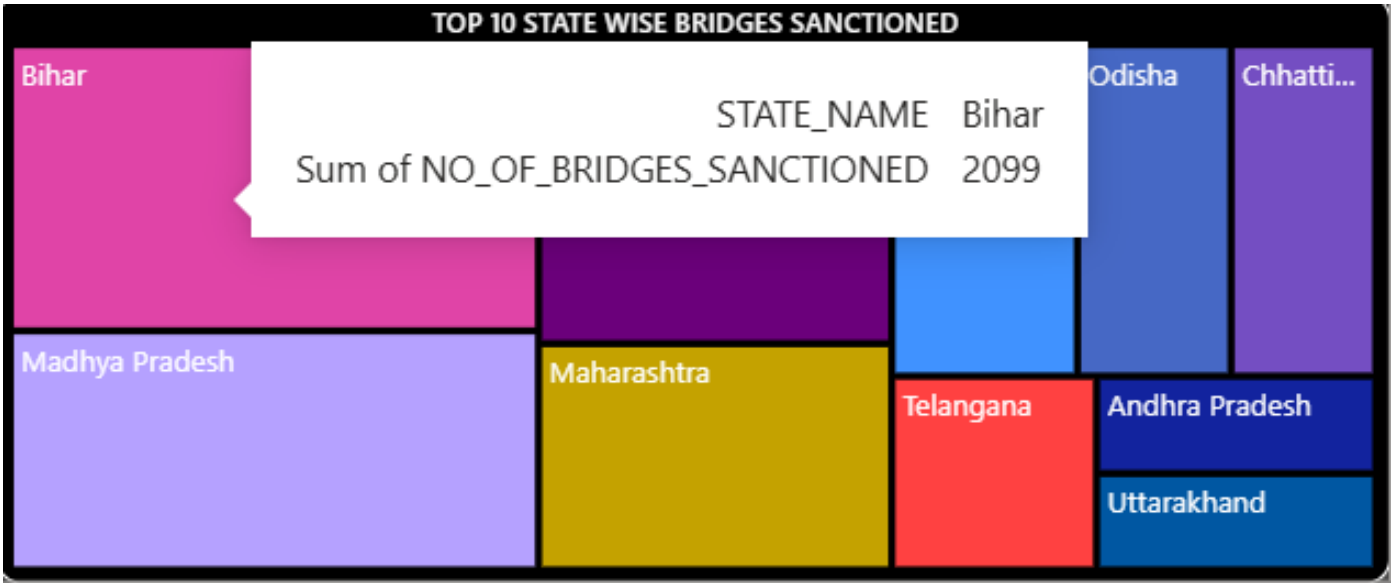
Performance Interpretation

State-wise comparative analysis not only highlights differences in overall expenditure and road length but also reveals important variations in completion percentage and expenditure utilization across states. These variations provide deeper insights into how effectively projects are being executed and managed at the state level.

Some states demonstrate a balanced level of physical and financial progress, where expenditure utilization closely aligns with road length completion. This balance indicates efficient planning, timely fund release, effective contractor management, and strong administrative coordination. Such states reflect good governance practices and can be considered as role models for effective PMGSY implementation.

In contrast, several states show noticeable gaps between sanctioned projects and completed works. In these cases, although a large number of projects or high financial allocations have been approved, the actual completion on the ground remains limited. These gaps may arise due to factors such as project delays, land acquisition issues, difficult terrain, weather conditions, or administrative bottlenecks. Identifying such patterns is essential for corrective action and improved monitoring.

This comparative analysis helps in identifying high-performing states that can serve as benchmarks for best practices, as well as low-performing states that require targeted intervention and support. By understanding these performance differences, policymakers can focus on improving implementation efficiency in lagging regions while replicating successful strategies from better-performing states.



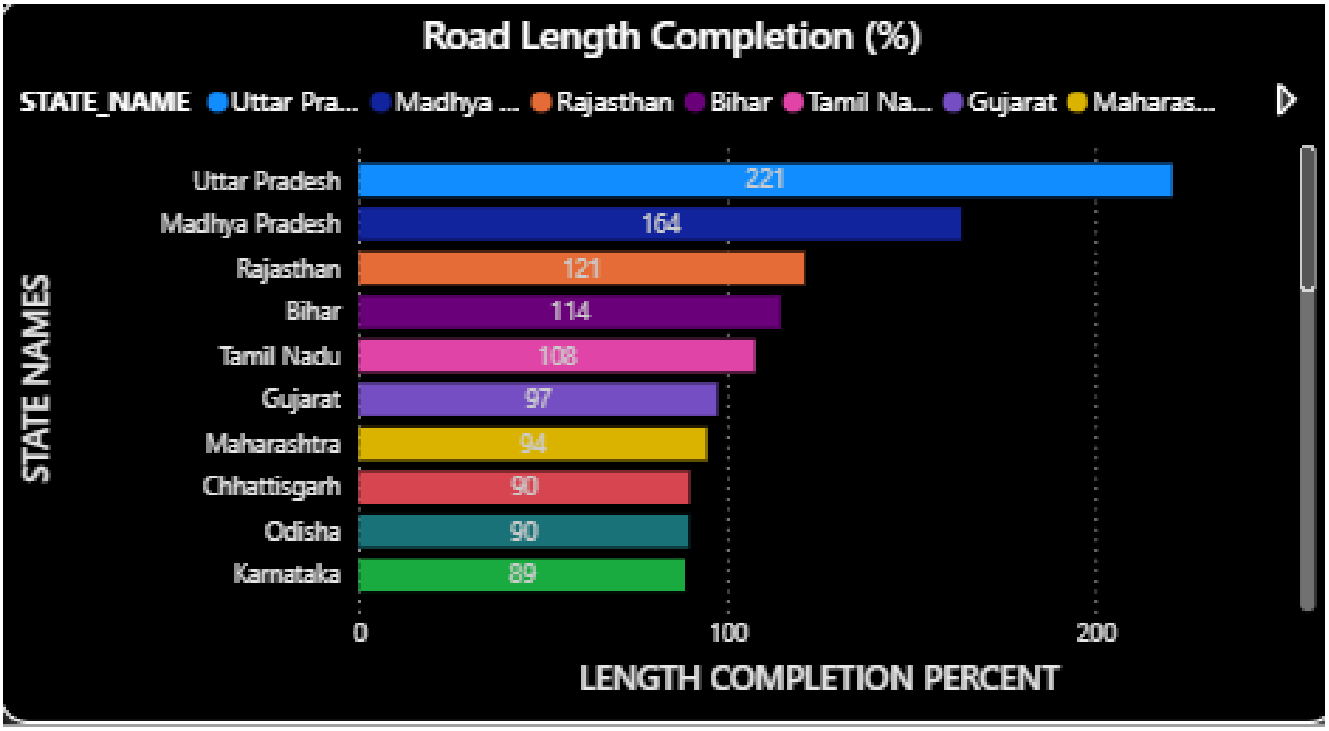
Significance of State-wise Analysis

State-wise analysis plays a crucial role in policy formulation, planning, and implementation of large-scale infrastructure programs such as PMGSY. Since each state has unique geographical, economic, and administrative conditions, a uniform approach to planning and execution may not always be effective. State-wise insights allow policymakers to adopt region-specific strategies that address local challenges more efficiently.

Understanding regional disparities helps in allocating resources more effectively and ensuring that public funds are utilized where they are most needed. For example, states with difficult terrain or remote locations may require higher investment per kilometer, while states with strong execution capacity may benefit from accelerated funding and expanded project scope.

State-wise comparative analysis also strengthens transparency and accountability. By clearly showing how different states perform in terms of road completion, expenditure utilization, and cost efficiency, it enables better performance tracking and evaluation. This promotes responsible use of public funds and encourages states to improve their execution standards.

Overall, state-wise analysis enhances performance monitoring and evidence-based decision-making within PMGSY. It supports continuous improvement by identifying strengths, weaknesses, and opportunities for intervention. As a result, such analysis contributes to more effective implementation of rural road infrastructure projects and helps achieve the long-term goal of inclusive and sustainable rural development.



7.5 State-wise Length Completion Percentage Analysis

Introduction

This analysis focuses on comparing the road length completion percentage across different states. Completion percentage is an important indicator of execution efficiency, as it reflects how much of the sanctioned road length has been successfully completed on the ground.

Description of the Visual

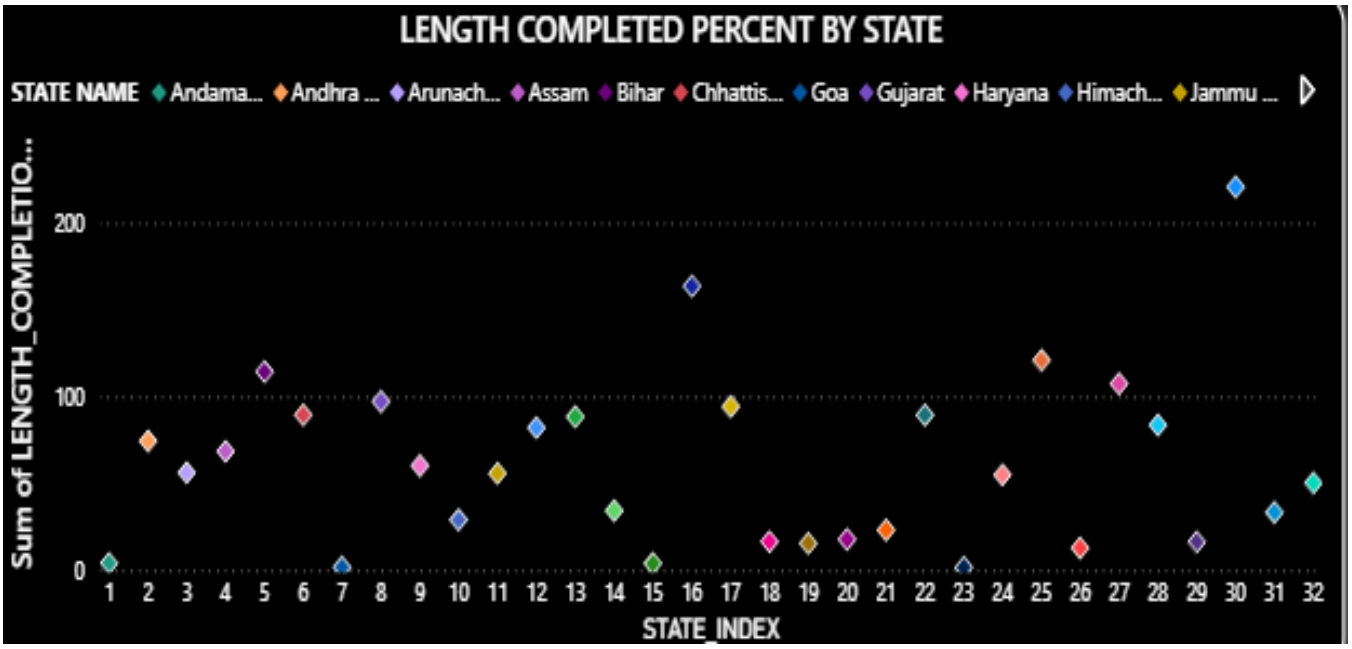
The scatter plot titled "Length Completed Percent by State" represents each state as a data point, plotted against a state index. The vertical axis shows the sum of length completion percentage, allowing easy comparison of performance across states.

Analysis and Interpretation

The visualization highlights significant variation in completion percentages among states. Some states exhibit very high completion percentages, indicating strong execution capacity and timely project implementation. These states demonstrate effective coordination between planning, funding, and on-ground execution.

On the other hand, several states show lower completion percentages, even though projects have been sanctioned. This indicates possible challenges such as delays in execution, administrative bottlenecks, difficult terrain, or contractor-related issues. The wide spread of points in the scatter plot clearly reflects uneven physical progress across regions.

Overall, this analysis helps in identifying high-performing states as benchmarks and low-performing states that require closer monitoring and intervention.



7.6 Expenditure vs Road Length Analysis by Scheme

Introduction

This section analyzes the relationship between financial expenditure and road length completion across different PMGSY schemes. Understanding this relationship is important to evaluate whether higher spending leads to proportionate physical outcomes.

Description of the Visual

The chart titled "Expenditure vs Length" compares two measures:

- Expenditure Occurred (in lakhs)
- Road Length Completion

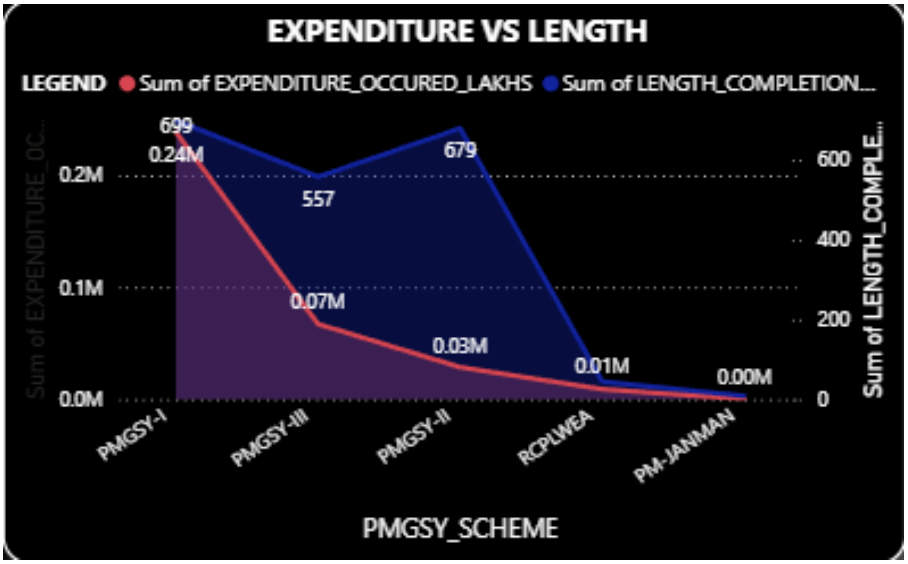
across PMGSY schemes such as PMGSY-I, PMGSY-II, PMGSY-III, RCPLWEA, and PM-JANMAN.

Analysis and Interpretation

The visual clearly shows that higher expenditure does not always result in higher road length completion. For example, some schemes show substantial expenditure but comparatively lower completed road length, indicating inefficiencies or execution delays.

PMGSY-I and PMGSY-III demonstrate stronger alignment between expenditure and physical output, reflecting better implementation efficiency. In contrast, smaller schemes such as RCPLWEA and PM-JANMAN show lower values for both expenditure and completed length, as they are targeted schemes with limited scope.

This analysis emphasizes the importance of monitoring both financial and physical indicators together, rather than evaluating performance based only on expenditure.



8. DASHBOARD DESIGN AND INTERACTIVITY

The Power BI dashboard developed in this project was designed with a strong emphasis on clarity, usability, and professional presentation. Since the target users of the dashboard include planners, administrators, and policymakers who may not have a technical background in data analytics, the design prioritizes simplicity and ease of interpretation. The primary objective of the dashboard is to transform complex PMGSY data into an intuitive visual format that supports quick understanding and informed decision-making.

Special attention was given to ensuring that the dashboard provides both high-level summaries and detailed insights. Key performance indicators (KPIs) are prominently displayed to give users an immediate overview of physical and financial progress, while detailed charts and visuals allow deeper exploration of the data. This layered design approach ensures that users can access the required level of detail without being overwhelmed by information.

8.1 Use of Interactive Slicers and Filters

To enhance interactivity, slicers were provided for important dimensions such as state, district, and scheme. These slicers allow users to dynamically filter the data and view performance for a specific region or PMGSY scheme. For example, a user can select a particular state to instantly view its sanctioned road length, completion percentage, expenditure utilization, and bridge construction status.

As slicer selections are changed, all visuals on the dashboard update automatically. This dynamic behavior ensures a smooth and responsive user experience, allowing users to explore multiple scenarios without refreshing or reloading the dashboard. Such interactivity makes the dashboard suitable for exploratory analysis and comparison across regions and schemes.

8.2 Drill-Down and Hierarchical Navigation

The dashboard also supports drill-down functionality, which enables users to move from a summarized view to more detailed levels of analysis. For instance, users can begin with a national-level overview and then drill down to state-level or scheme-level performance. This hierarchical navigation helps users understand both the overall picture and region-specific details.

Drill-down features are particularly useful in identifying problem areas. If a particular state shows low completion or utilization rates, users can drill down further to analyze scheme-wise or district-level performance. This feature supports deeper investigation without cluttering the dashboard with excessive visuals.

8.3 Cross-Filtering and Cross-Highlighting

To further improve analytical capability, cross-filtering and cross-highlighting features were enabled across all visuals. This means that selecting a data point in one chart automatically

highlights related data in other visuals. For example, selecting a state in a bar chart will instantly update the map, treemap, and KPI cards to reflect that state's performance.

This interconnected behavior enhances the exploratory nature of the dashboard and helps users identify relationships between physical progress and financial indicators. It also ensures consistency across visuals and improves overall data interpretation.

8.4 Visual Design and Theme Selection

A dark theme was chosen for the dashboard to improve readability and provide a professional appearance. Dark backgrounds reduce visual strain, especially when dashboards are viewed for extended periods. Contrasting colors were carefully selected so that important KPIs, charts, and values stand out clearly against the background.

Consistent color usage was maintained across similar metrics to avoid confusion. For example, completion-related metrics use similar color shades, while expenditure-related metrics follow a different color pattern. This consistency helps users quickly associate visuals with specific performance indicators.

Font style, size, alignment, and spacing were kept uniform throughout the dashboard. Proper spacing between visuals prevents clutter and improves readability. Clear titles, axis labels, and legends were added to ensure that each visual is self-explanatory.

8.5 Selection of Visuals

The dashboard uses a combination of KPI cards, bar charts, maps, treemaps, waterfall charts, and scatter plots. Each visual was selected based on its suitability for representing specific data:

- KPI cards provide a quick snapshot of key metrics
- Bar and column charts enable state-wise and scheme-wise comparisons
- Maps highlight geographical variations in cost and performance
- Treemaps show dominance and contribution of top-performing states
- Waterfall charts explain scheme-wise cost distribution

This thoughtful selection of visuals ensures that data is presented in the most effective and meaningful way.

8.6 User Experience and Decision Support

From a user experience perspective, the dashboard was designed to minimize complexity while maximizing insight. Users do not need prior knowledge of Power BI to navigate the dashboard. Clear visual cues, intuitive slicers, and interactive elements guide users naturally through the analysis.

Overall, the dashboard functions as an effective decision-support tool. It enables stakeholders to quickly identify performance gaps, monitor progress, compare states and schemes, and evaluate cost efficiency. By presenting both physical and financial indicators in an integrated and interactive manner, the dashboard supports better planning, improved monitoring, and evidence-based decision-making.

In conclusion, the dashboard design and interactivity play a crucial role in the success of this project. By combining strong visual design principles with powerful interactive features, the Power BI dashboard effectively transforms PMGSY data into actionable insights that can support governance and infrastructure development.

9. KEY INSIGHTS AND FINDINGS

- The analysis of PMGSY data using the Power BI dashboard resulted in several important insights related to both physical progress and financial performance. These insights help in understanding how effectively rural road infrastructure projects are being implemented across different states and schemes.
- One of the key findings is that higher expenditure does not always result in higher project completion rates. In some states and schemes, large amounts of funds have been sanctioned and spent, yet the road length completion percentage remains relatively low. This indicates the presence of implementation challenges such as delays, administrative issues, or geographical constraints that affect timely execution.
- Another important insight is that the cost per kilometer of road construction varies significantly across states. Plains and easily accessible regions generally show lower construction costs, while hilly, forested, and remote regions—particularly in northeastern states—have much higher costs. This variation highlights the influence of terrain, climate, and accessibility on infrastructure development.
- The analysis also reveals that bridge construction plays a critical role in improving rural connectivity, especially in regions with rivers, uneven terrain, and seasonal accessibility issues. States with higher numbers of completed bridges tend to show better connectivity outcomes, even if total road length is comparatively lower.
- Finally, scheme-wise allocation patterns reflect government policy priorities. Schemes such as PMGSY-I and PMGSY-III receive higher financial allocations, indicating a strong focus on both expanding rural road networks and upgrading existing infrastructure. Newer schemes, although smaller in scale, target specific regions and development challenges, demonstrating a focused and inclusive policy approach.
- Overall, these insights provide valuable inputs for policymakers and planners to improve project execution, allocate resources efficiently, and strengthen rural infrastructure planning.
- In addition to these findings, the integrated analysis of physical and financial indicators highlights the importance of continuous monitoring and data-driven evaluation in large-scale infrastructure programs. The Power BI dashboard enables timely identification of performance gaps and supports proactive decision-making. Such insights can help improve transparency, accountability, and overall effectiveness in the implementation of PMGSY projects.

10. CONCLUSION

This project successfully demonstrates the effective use of Power BI as a data analytics and visualization tool to analyze large-scale government infrastructure data related to the Pradhan Mantri Gram Sadak Yojana (PMGSY). By integrating physical progress and financial performance data, the project provides a comprehensive and meaningful view of rural road development across India.

One of the major strengths of this project is its ability to transform complex and large datasets into clear, interactive, and easy-to-understand dashboards. Traditional tabular reports often fail to highlight patterns and performance gaps. In contrast, the Power BI dashboards developed in this project enable stakeholders to quickly identify trends, compare state-wise and scheme-wise performance, and monitor progress at multiple levels. This improves transparency and supports informed decision-making.

The physical progress analysis highlights significant variation in road construction and bridge completion across states. While some states demonstrate high completion percentages and efficient execution, others show delays despite having a large number of sanctioned projects. This indicates that factors such as administrative efficiency, terrain conditions, and project management practices play a crucial role in determining physical progress.

Similarly, the financial performance analysis reveals important insights regarding expenditure utilization. The study shows that higher financial allocation does not always result in better project outcomes. In several cases, sanctioned funds remain underutilized, pointing towards implementation challenges such as delays, planning issues, or coordination gaps. The analysis of cost per kilometer further emphasizes how geographical conditions influence construction costs, especially in hilly and remote regions.

The scheme-wise analysis provides clarity on government policy priorities. Older schemes like PMGSY-I and development-focused schemes like PMGSY-III account for the majority of investment, reflecting a balanced approach towards both expanding and upgrading rural road networks. Meanwhile, region-specific schemes, although smaller in scale, play an important role in addressing connectivity issues in difficult and underserved areas.

State-wise comparative analysis offers valuable insights into regional disparities in infrastructure development. States with large rural populations naturally receive higher investments, while northeastern and mountainous states face higher construction costs due to challenging terrain. This highlights the importance of adopting region-specific planning and execution strategies rather than a uniform approach.

From a technical perspective, this project also demonstrates strong skills in data cleaning, modeling, and analysis. The use of Power Query for data transformation, DAX for calculating key performance indicators, and a structured data model ensures accuracy and reliability of results.

Interactive features such as slicers, drill-downs, and cross-filtering enhance user engagement and make the dashboard a powerful exploratory tool.

Overall, this project proves that data analytics and visualization can play a crucial role in monitoring and improving government infrastructure programs. By presenting actionable insights in a simple and visual manner, the Power BI dashboard developed in this project can support policymakers, planners, and administrators in improving planning efficiency, resource allocation, and project execution. The approach followed in this study can also be extended to other large-scale public sector projects, making it a valuable contribution to data-driven governance.

11. FUTURE SCOPE

Although this project provides a detailed and meaningful analysis of the physical and financial performance of PMGSY projects, there is still significant potential to enhance and extend the scope of this work in the future. With continuous advancements in data analytics technologies and increasing availability of digital government data, dashboards like this can be made more powerful, dynamic, and decision-oriented.

One important area for future improvement is the integration of real-time data updates. In the current project, analysis is performed using static or periodically updated datasets. In the future, the dashboard can be connected directly to live data sources from government monitoring systems. This would allow real-time tracking of project progress, enabling authorities to monitor construction status, expenditure, and delays on a daily or weekly basis. Real-time insights can support faster decision-making and timely corrective actions.

Another promising extension is the use of predictive analysis to identify potential project delays. By analyzing historical PMGSY data, predictive models can be developed to estimate the likelihood of delays based on factors such as expenditure patterns, project size, location, and past performance. Early identification of delay-prone projects would help planners and administrators take preventive steps, such as reallocating resources or improving coordination.

The scope of the project can also be expanded to include district-level or block-level micro analysis. While state-level analysis provides a broad overview, district-level insights can reveal local challenges more accurately. This deeper level of analysis would be particularly useful for local authorities to address region-specific issues such as terrain difficulties, contractor performance, or funding constraints.

Finally, machine learning-based performance prediction can be incorporated to evaluate future outcomes of PMGSY projects. Machine learning models can analyze multiple variables such as expenditure trends, geographical conditions, scheme type, and historical completion rates to predict future performance indicators. Such predictions can support long-term planning and policy formulation by providing data-driven forecasts.

12. REFERENCES

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13. LINKEDIN

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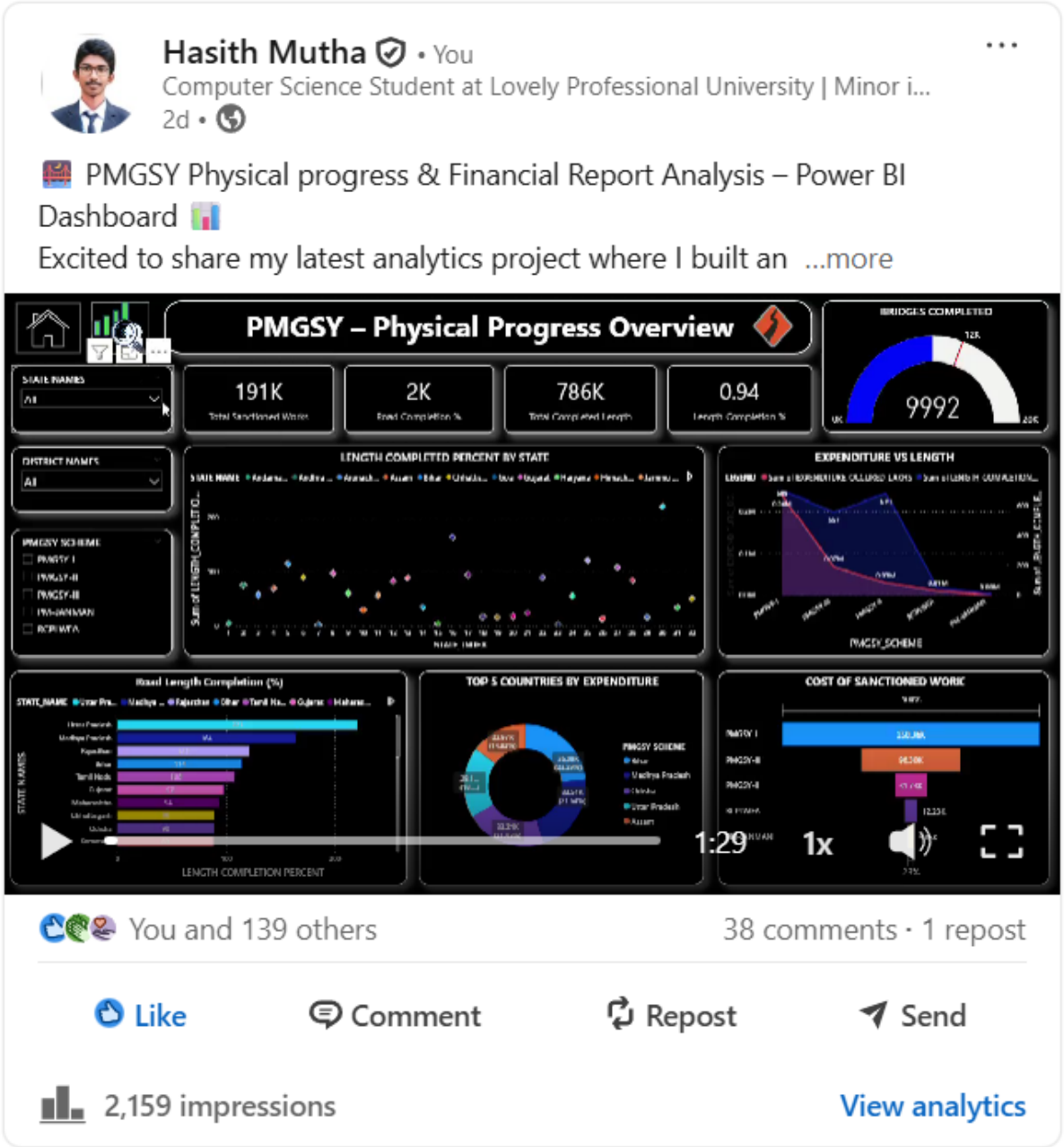
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